

Q1. In an expression $a \times 10^b$;

- (1) b is order of magnitude for $a \geq 5$ (2) b is order of magnitude for $a \leq 5$
 (3) a is order of magnitude for $b \leq 5$ (4) b is order of magnitude for $5 < a \leq 10$

Q2. Young's modulus is determined by the equation given by $Y = \frac{49000}{l} \frac{M}{\text{cm}^2}$ where M is the mass and l is the extension of wire used in the experiment. Now error in Young modulus (Y) is estimated by taking data from $M - l$ plot in graph paper. The smallest scale divisions are 5 g and 0.02 cm along load axis and extension axis respectively. If the value of M and l are 500 g and 2 cm respectively then percentage error of Y is :

- (1) 0.5% (2) 2%
 (3) 0.02% (4) 0.2%

Q3. A clock has 75 cm, 60 cm long second hand and minute hand respectively. In 30 minutes duration the tip of second hand will travel x distance more than the tip of minute hand. The value of x in meter is nearly (Take $\pi = 3.14$) :

- (1) 140.5 (2) 118.9
 (3) 139.4 (4) 220.0

Q4. A stationary particle breaks into two parts of masses m_A and m_B which move with velocities v_A and v_B respectively. The ratio of their kinetic energies ($K_B : K_A$) is :

- (1) $v_B : v_A$ (2) $m_B : m_A$
 (3) $m_B v_B : m_A v_A$ (4) 1 : 1

Q5. Three bodies A, B and C have equal kinetic energies and their masses are 400 g, 1.2 kg and 1.6 kg respectively. The ratio of their linear momenta is :

- (1) $\sqrt{2} : \sqrt{3} : 1$ (2) $1 : \sqrt{3} : 2$
 (3) $1 : \sqrt{3} : \sqrt{2}$ (4) $\sqrt{3} : \sqrt{2} : 1$

Q6. A player caught a cricket ball of mass 150 g moving at a speed of 20 m/s. If the catching process is completed in 0.1 s, the magnitude of force exerted by the ball on the hand of the player is:

- (1) 3 N (2) 300 N
 (3) 150 N (4) 30 N

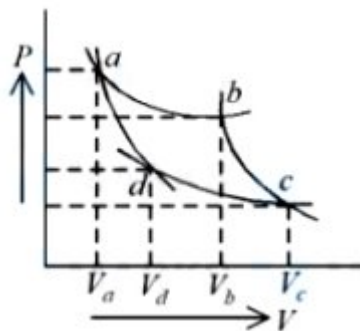
Q7. Two planets A and B having masses m_1 and m_2 move around the sun in circular orbits of r_1 and r_2 radii respectively. If angular momentum of A is L and that of B is $3L$, the ratio of time period $\left(\frac{T_A}{T_B}\right)$ is:

- (1) $\left(\frac{r_2}{r_1}\right)^{\frac{3}{2}}$ (2) $\frac{1}{27} \left(\frac{m_2}{m_1}\right)^3$
 (3) $27 \left(\frac{m_1}{m_2}\right)^3$ (4) $\left(\frac{r_1}{r_2}\right)^3$

Q8. Correct Bernoulli's equation is (symbols have their usual meaning) :

- (1) $P + mgh + \frac{1}{2}mv^2 = \text{constant}$ (2) $P + \rho gh + \frac{1}{2}\rho v^2 = \text{constant}$
 (3) $P + \rho gh + \rho v^2 = \text{constant}$ (4) $P + \frac{1}{2}\rho gh + \frac{1}{2}\rho v^2 = \text{constant}$

Q9. Two different adiabatic paths for the same gas intersect two isothermal curves as shown in P-V diagram. The



relation between the ratio $\frac{V_a}{V_d}$ and the ratio $\frac{V_b}{V_c}$ is:

(1) $\frac{V_a}{V_d} \neq \frac{V_b}{V_c}$

(2) $\frac{V_a}{V_d} = \frac{V_b}{V_c}$

(3) $\frac{V_a}{V_d} = \left(\frac{V_b}{V_c}\right)^{-1}$

(4) $\frac{V_a}{V_d} = \left(\frac{V_b}{V_c}\right)^2$

Q10. A mixture of one mole of monoatomic gas and one mole of a diatomic gas (rigid) are kept at room temperature (27°C). The ratio of specific heat of gases at constant volume respectively is:

(1) $\frac{7}{5}$

(2) $\frac{3}{5}$

(3) $\frac{5}{3}$

(4) $\frac{3}{2}$

Q11. Two charged conducting spheres of radii a and b are connected to each other by a conducting wire. The ratio of charges of the two spheres respectively is:

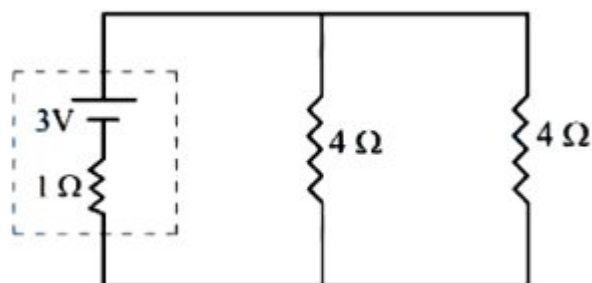
(1) $\frac{a}{b} v$

(2) $\sqrt{ab}$

(3) $\frac{b}{a}$

(4) ab

Q12. In the given circuit, the terminal potential difference of the cell is :



(1) 2 V

(2) 3 V

(3) 4 V

(4) 1.5 V

Q13. Paramagnetic substances: A. align themselves along the directions of external magnetic field. B. attract strongly towards external magnetic field. C. has susceptibility little more than zero. D. move from a region of strong magnetic field to weak magnetic field. Choose the most appropriate answer from the options given below:

(1) A, B, C Only

(2) A, B, C, D

(3) A, C Only

(4) B, D Only

Q14. A LCR circuit is at resonance for a capacitor C , inductance L and resistance R . Now the value of resistance is halved keeping all other parameters same. The current amplitude at resonance will be now:

- (1) Zero (2) same
(3) halved (4) double

Q15. Critical angle of incidence for a pair of optical media is 45° . The refractive indices of first and second media are in the ratio:

- (1) $1 : \sqrt{2}$ (2) $\sqrt{2} : 1$
(3) $2 : 1$ (4) $1 : 2$

Q16. A proton and an electron are associated with same de-Broglie wavelength. The ratio of their kinetic energies is:

(Assume $h = 6.63 \times 10^{-34}$ J s, $m_e = 9.0 \times 10^{-31}$ kg and $m_p = 1836$ times m_e)

- (1) $1 : \sqrt{1836}$ (2) $1 : \frac{1}{1836}$
(3) $1 : \frac{1}{\sqrt{1836}}$ (4) $1 : 1836$

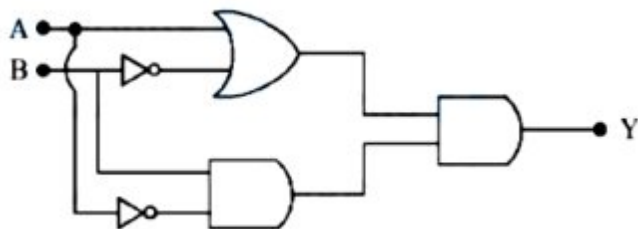
Q17. Average force exerted on a non-reflecting surface at normal incidence is 2.4×10^{-4} N. If 360 W/cm^2 is the light energy flux during span of 1 hour 30 minutes, Then the area of the surface is:

- (1) 0.2 m^2 (2) 20 m^2
(3) 0.1 m^2 (4) 0.02 m^2

Q18. Binding energy of a certain nucleus is 18×10^8 J. How much is the difference between total mass of all the nucleons and nuclear mass of the given nucleus:

- (1) $10 \mu\text{g}$ (2) $20 \mu\text{g}$
(3) $0.2 \mu\text{g}$ (4) $2 \mu\text{g}$

Q19. The output Y of following circuit for given inputs is :

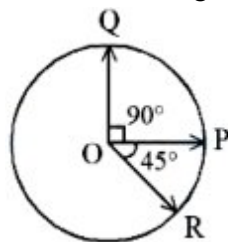


- (1) $A \cdot B(A + B)$ (2) 0
(3) $\overline{A} \cdot B$ (4) $A \cdot B$

Q20. The diameter of a sphere is measured using a vernier caliper whose 9 divisions of main scale are equal to 10 divisions of vernier scale. The shortest division on the main scale is equal to 1 mm. The main scale reading is 2 cm and second division of vernier scale coincides with a division on main scale. If mass of the sphere is 8.635 g, the density of the sphere is:

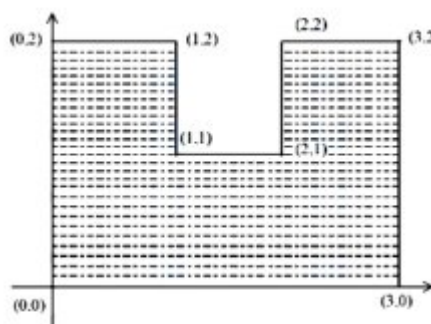
- (1) 2.0 g/cm^3 (2) 1.7 g/cm^3
(3) 2.2 g/cm^3 (4) 2.5 g/cm^3

- Q21. Three vectors $\vec{OP}$, $\vec{OQ}$ and $\vec{OR}$ each of magnitude A are acting as shown in figure. The resultant of the three



vectors is $A\sqrt{x}$. The value of x is _____.

- Q22. A uniform thin metal plate of mass 10 kg with dimensions is shown. The ratio of x and y coordinates of center



of mass of plate in $\frac{n}{9}$. The value of n is _____

- Q23. A liquid column of height 0.04 cm balances excess pressure of a soap bubble of certain radius. If density of liquid is $8 \times 10^3 \text{ kg m}^{-3}$ and surface tension of soap solution is 0.28 Nm^{-1} , then diameter of the soap bubble is _____ cm. (if $g = 10 \text{ m s}^{-2}$)

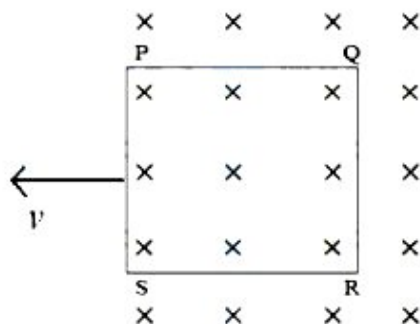
- Q24. A closed and an open organ pipe have same lengths. If the ratio of frequencies of their seventh overtones is $\left(\frac{a-1}{a}\right)$ then the value of a is _____

- Q25. An electric field, $\vec{E} = \frac{2\hat{i} + 6\hat{j} + 8\hat{k}}{\sqrt{6}}$ passes through the surface of 4 m^2 area having unit vector $\hat{n} = \left(\frac{2\hat{i} + \hat{j} + \hat{k}}{\sqrt{6}}\right)$. The electric flux for that surface is _____ Vm.

- Q26. Resistance of a wire at 0°C , 100°C and $t^\circ\text{C}$ is found to be 10Ω , 10.2Ω and 10.95Ω respectively. The temperature t in Kelvin scale is _____

- Q27. An electron with kinetic energy 5eV enters a region of uniform magnetic field of $3 \mu\text{T}$ perpendicular to its direction. An electric field E is applied perpendicular to the direction of velocity and magnetic field. The value of E , so that electron moves along the same path, is _____ NC^{-1} . (Given, mass of electron = $9 \times 10^{-31} \text{ kg}$, electric charge = $1.6 \times 10^{-19} \text{ C}$)

- Q28. A square loop PQRS having 10 turns, area $3.6 \times 10^{-3} \text{ m}^2$ and resistance 100Ω is slowly and uniformly being pulled out of a uniform magnetic field of magnitude $B = 0.5 \text{ T}$ as shown. Work done in pulling the loop out of



the field in 1.0 s is _____ $\times 10^{-6}$ J.

Q29. A parallel beam of monochromatic light of wavelength 600 nm passes through single slit of 0.4 mm width. Angular divergence corresponding to second order minima would be _____ $\times 10^{-3}$ rad.

Q30. In an alpha particle scattering experiment distance of closest approach for the α particle is 4.5×10^{-14} m. If target nucleus has atomic number 80, then maximum velocity of α - particle is _____ $\times 10^5$ m/s approximately. $\left(\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ SI unit, mass of α particle = 6.72×10^{-27} kg)

Q31. Combustion of glucose ($\text{C}_6\text{H}_{12}\text{O}_6$) produces CO_2 and water. The amount of oxygen (in g) required for the complete combustion of 900 g of glucose is : [Molar mass of glucose in $\text{gmol}^{-1} = 180$]

(1) 480

(2) 800

(3) 960

(4) 32

Q32.

Match List I with List II

	List - I		List - II
	(Elements)		(Properties in their respective groups)
A.	Cl, S	I.	Elements with highest electronegativity
B.	Ge, As	II.	Elements with largest atomic size
C.	Fr, Ra	III.	Elements which show properties of both metals and non-metal
D.	F, O	IV.	Elements with highest negative electron gain enthalpy

Choose

the correct answer from the options given below:

(1) A-II, B-I, C-IV, D-III

(2) A-III, B-II, C-I, D-IV

(3) A-IV, B-III, C-II, D-I

(4) A-II, B-III, C-IV, D-I

Q33.

Match List I with List II

	List-I		List-II
	(Molecule)		(Shape)
A.	NH_3	I.	Square pyramid
B.	BrF_5	II.	Tetrahedral
C.	PCl_5	III.	Trigonal pyramidal
D.	CH_4	IV.	Trigonal bipyramidal

Choose the correct answer

from the options given below:

(1) A-II, B-IV, C-I, D-III

(2) A-III, B-I, C-IV, D-II

(3) A-IV, B-III, C-I, D-II

(4) A-III, B-IV, C-I, D-II

Q34.

Match List I with List II

	List-I		List-II
	(Molecule)		(Shape)
A.	$\text{Fe}_4[\text{Fe}(\text{CN})_6]_3 \cdot x\text{H}_2\text{O}$	I.	Violet
B.	$[\text{Fe}(\text{CN})_5\text{NOS}]^{4-}$	II.	Blood Red A
C.	$[\text{Fe}(\text{SCN})]^{2+}$	III.	Prussian Blue
D.	$(\text{NH}_4)_3\text{PO}_4 \cdot 12\text{MoO}_3$	IV.	Yellow

Choose the correct answer

from the options given below:

(1) A-III, B-I, C-II, D-IV

(2) A-I, B-II, C-III, D-IV

(3) A-IV, B-I, C-II, D-III

(4) A-II, B-III, C-IV, D-I

Q35. Given below are two statements: Statement I: $\text{N}(\text{CH}_3)_3$ and $\text{P}(\text{CH}_3)_3$ can act as ligands to form transition metal complexes. Statement II: As N and P are from same group, the nature of bonding of $\text{N}(\text{CH}_3)_3$ and $\text{P}(\text{CH}_3)_3$ is always same with transition metals. In the light of the above statements, choose the most appropriate answer from the options given below:

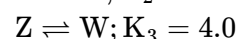
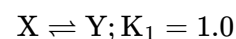
(1) Statement I is correct but Statement II is incorrect.

(2) Statement I is incorrect but Statement II is correct.

(3) Both Statement I and Statement II are correct.

(4) Both Statement I and Statement II are incorrect.

Q36.

For the given hypothetical reactions, the equilibrium constants are as follows : $\text{Y} \rightleftharpoons \text{Z}; K_2 = 2.0$ Theequilibrium constant for the reaction $\text{X} \rightleftharpoons \text{W}$ is

(1) 6.0

(2) 12.0

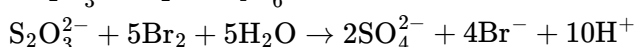
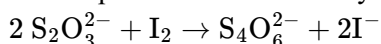
(3) 7.0

(4) 8.0

Q37. Among the following halogens F_2 , Cl_2 , Br_2 and I_2 Which can undergo disproportionation reactions?

(1) F_2 , Cl_2 and Br_2 (2) F_2 and Cl_2 (3) Only I_2 (4) Cl_2 , Br_2 and I_2

Q38. Thiosulphate reacts differently with iodine and bromine in the reactions given below:



behaviour of thiosulphate?

(1) Bromine is a stronger oxidant than iodine

(2) Thiosulphate undergoes oxidation by bromine and reduction by iodine in these reaction

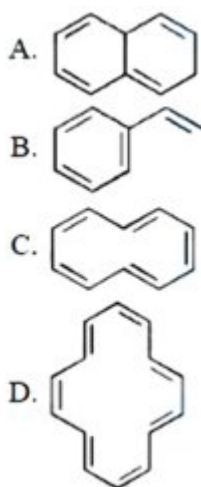
(3) Bromine is a weaker oxidant than iodine

(4) Bromine undergoes oxidation and iodine undergoes reduction in these reactions

Q39. Give below are two statements: One is labelled as Assertion A and the other is labelled as Reason R: Assertion A: The stability order of +1 oxidation state of Ga, In and Tl is $\text{Ga} < \text{In} < \text{Tl}$. Reason R: The inert pair effect stabilizes the lower oxidation state down the group. In the light of the above statements, choose the correct answer from the options given below:

- (1) A is true but R is false. (2) A is false but R is true.
 (3) Both A and R are true and R is the correct explanation of A. (4) Both A and R are true but R is NOT the correct explanation of A.

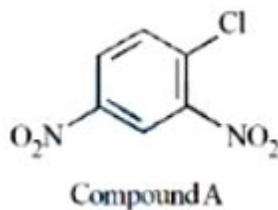
Q40.



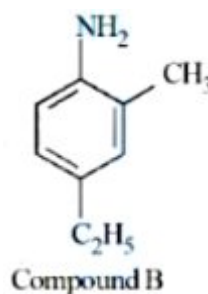
Which of the following are aromatic?

- (1) A and C only (2) B and D only
 (3) C and D only (4) A and B only

Q41.



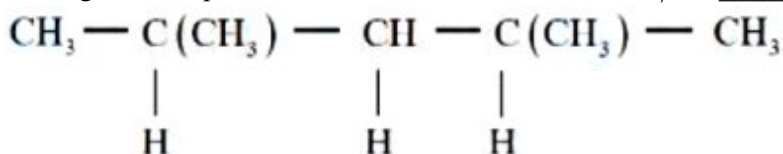
Given below are two statements: Statements I:



IUPAC name of Compound A is 4-chloro-1,3-dinitrobenzene. Statements II:

IUPAC name of Compound B is 4-ethyl-2-methylaniline. In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both Statement I and Statement II are incorrect. (2) Both Statement I and Statement II are correct.
 (3) Statement I is correct but Statement II is incorrect. (4) Statement I is incorrect but Statement II is correct.

Q42. In the given compound, the number of 2° carbon atom /s is _____.

- (1) Four (2) Two
 (3) One (4) Three

Q43. Iron (III) catalyses the reaction between iodide and persulphate ions, in which A. Fe^{3+} oxidises the iodide ion
 B. Fe^{3+} oxidises the persulphate ion C. Fe^{2+} reduces the iodide ion D. Fe^{2+} reduces the persulphate ion
 Choose the most appropriate answer from the options given below:

- (1) B only (2) A only
 (3) B and C only (4) A and D only

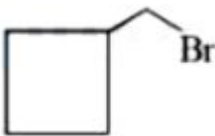
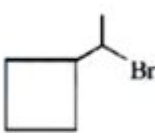
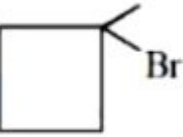
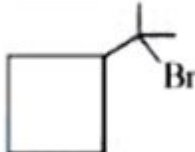
Q44. Number of Complexes with even number of electrons in t_{2g} orbitals is -
 $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$, $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$, $[\text{Co}(\text{H}_2\text{O})_6]^{3+}$, $[\text{Cu}(\text{H}_2\text{O})_6]^{2+}$, $[\text{Cr}(\text{H}_2\text{O})_6]^{2+}$

- (1) 2 (2) 3
 (3) 1 (4) 5

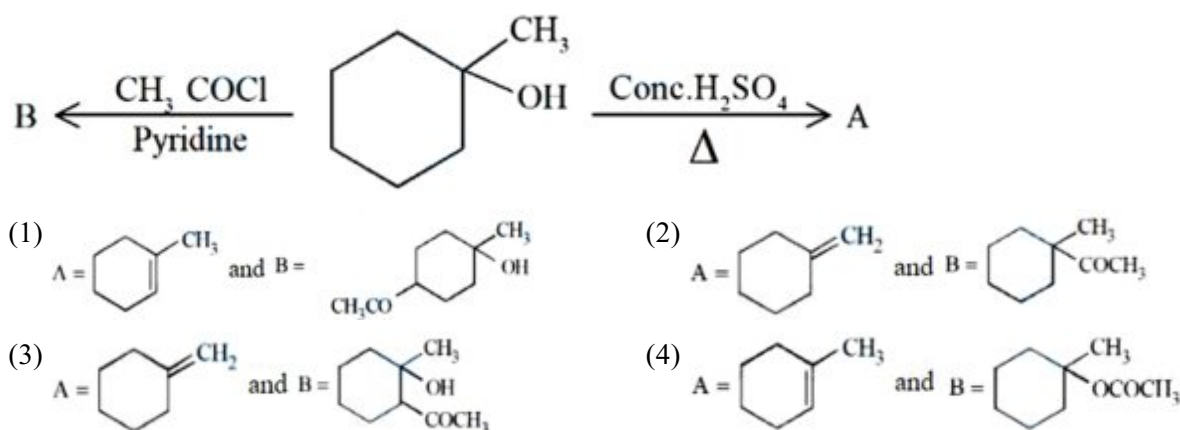
Q45. An octahedral complex with the formula $\text{CoCl}_3 \cdot n\text{NH}_3$ upon reaction with excess of AgNO_3 solution gives 2 moles of AgCl . Consider the oxidation state of Co in the complex is 'x'. The value of "x + n" is _____

- (1) 6 (2) 8
 (3) 3 (4) 5

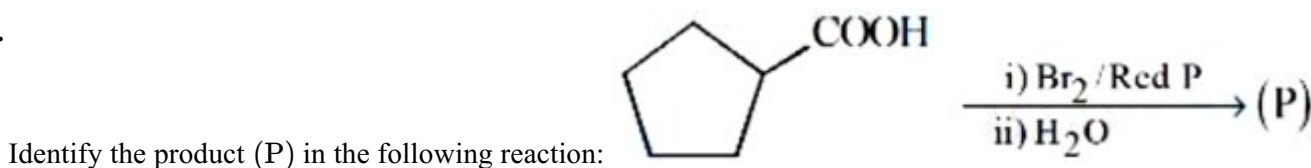
Q46. Which among the following compounds will undergo fastest $\text{S}_{\text{N}}2$ reaction.

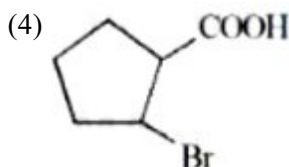
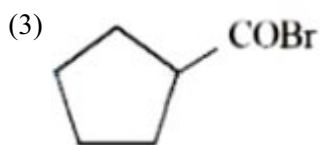
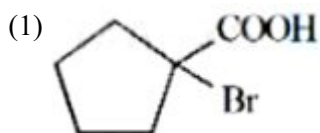
- (1)  (2) 
 (3)  (4) 

Q47. Identify the major products A and B respectively in the following set of reactions.



Q48.





Q49.

Match List I with List II

	List-I		List-II
	(Name of the test)		(Reaction sequence involved) [M is metal]
A.	Borax bead test	I.	$\text{MCO}_3 \rightarrow \text{MO} \xrightarrow{\text{Co(NO}_3)_2} \text{CoO} \cdot \text{MO}$
B.	Charcoal cavity test	II.	$\text{MCO}_3 \rightarrow \text{MCl}_2 \rightarrow \text{M}^{2+}$
C.	Cobalt nitrate test	III.	$\text{MSO}_4 \xrightarrow[\Delta]{\text{Na}_2\text{B}_4\text{O}_7} \text{M(BO}_2)_2 \rightarrow \text{MBO}_2 \rightarrow \text{M}$
D.	Flame test	IV.	$\text{MSO}_4 \xrightarrow[\Delta]{\text{Na}_2\text{CO}_3} \text{MCO}_3 \rightarrow \text{MO} \rightarrow \text{M}$

Choose the correct answer from the options given below:

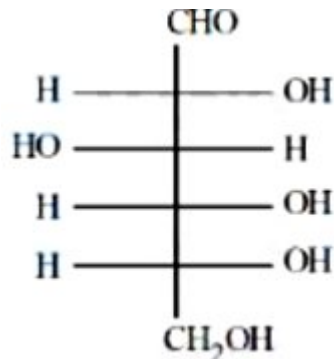
(1) A-III, B-II, C-IV, D-I

(2) A-III, B-IV, C-I, D-II

(3) A-III, B-I, C-II, D-IV

(4) A-III, B-I, C-IV, D-II

Q50.



The incorrect statement regarding the given structure is

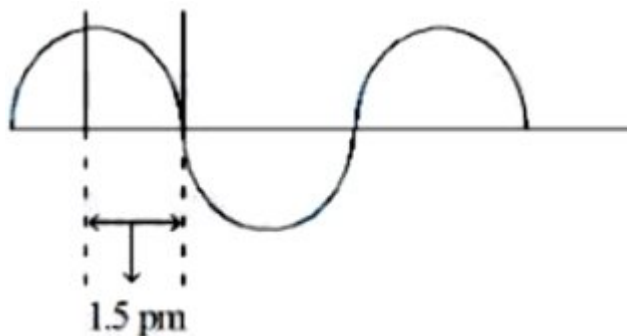
(1) can be oxidized to a dicarboxylic acid with Br_2 water

(2) will coexist in equilibrium with 2 other cyclic structure

(3) despite the presence of -CHO does not give Schiff's test

(4) has 4 asymmetric carbon atom

Q51.



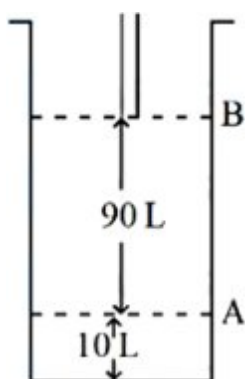
A hypothetical electromagnetic wave is shown below.

The frequency of the wave is $x \times 10^{19}$ Hz. $x =$ _____ (nearest integer)

Q52. Number of molecules from the following which are exceptions to octet rule is _____

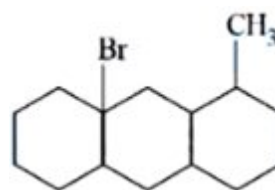
$\text{CO}_2, \text{NO}_2, \text{H}_2\text{SO}_4, \text{BF}_3, \text{CH}_4, \text{SiF}_4, \text{ClO}_2, \text{PCl}_5, \text{BeF}_2, \text{C}_2\text{H}_6, \text{CHCl}_3, \text{CBr}_4$

Q53.



Consider the figure provided. 1 mol of an ideal gas is kept in a cylinder, fitted with a piston, at the position A, at 18°C . If the piston is moved to position B, keeping the temperature unchanged, then ' x ' L atm work is done in this reversible process. $x =$ _____ L atm. (nearest integer) [Given : Absolute temperature $= ^\circ\text{C} + 273.15$, $R = 0.08206 \text{ L atm mol}^{-1} \text{ K}^{-1}$]

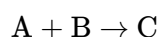
Q54.



The number of optical isomers in following compound is: _____

Q55. A solution containing 10 g of an electrolyte AB_2 in 100 g of water boils at 100.52°C . The degree of ionization of the electrolyte (α) is _____ $\times 10^{-1}$. (nearest integer) [Given : Molar mass of $\text{AB}_2 = 200 \text{ g mol}^{-1}$, K_b (molal boiling point elevation const. of water) $= 0.52 \text{ K kg mol}^{-1}$, boiling point of water $= 100^\circ\text{C}$; AB_2 ionises as $\text{AB}_2 \rightarrow \text{A}^{2+} + 2 \text{B}^-$]

Q56. Consider the following reaction

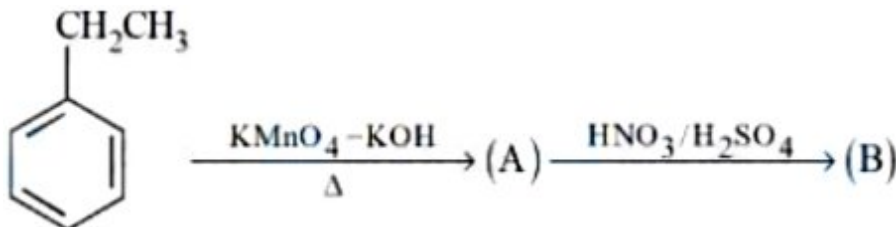


The time taken for A to become $1/4^{\text{th}}$ of its initial concentration is twice the time taken to become $1/2$ of the same. Also, when the change of concentration of B is plotted against time, the resulting graph gives a straight

line with a negative slope and a positive intercept on the concentration axis. The overall order of the reaction is _____

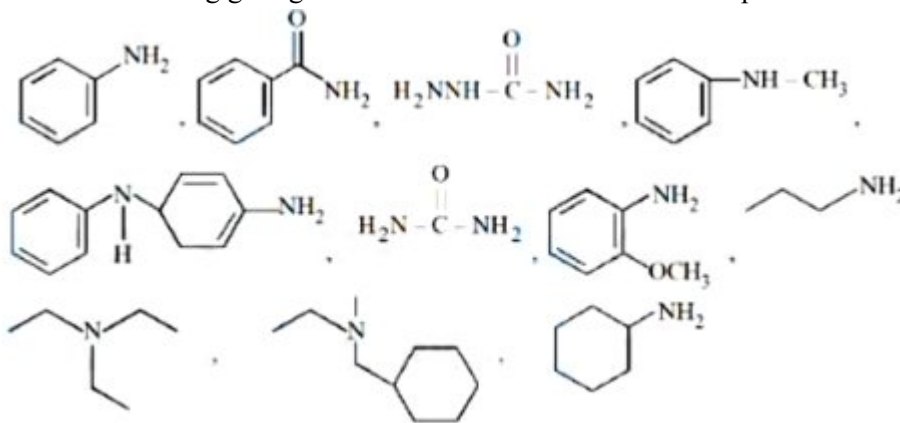
Q57. The 'spin only' magnetic moment value of MO_4^{2-} is _____ BM. (Where M is a metal having least metallic radii. among Sc, Ti, V, Cr, Mn and Zn). (Given atomic number: Sc = 21, Ti = 22, V = 23, Cr = 24, Mn = 25 and Zn = 30)

Q58. Major product *B* of the following reaction has _____ π -bond.



Q59. If 279 g of aniline is reacted with one equivalent of benzenediazonium chloride, the maximum amount of aniline yellow formed will be _____ g. (nearest integer) (consider complete conversion).

Q60. Number of amine compounds from the following giving solids which are soluble in NaOH upon reaction with



Hinsberg's reagent is _____

Q61. The sum of all the solutions of the equation $(8)^{2x} - 16 \cdot (8)^x + 48 = 0$ is :

- (1) $1 + \log_8(6)$ (2) $1 + \log_6(8)$
(3) $\log_8(6)$ (4) $\log_8(4)$

Q62. Let z be a complex number such that $|z + 2| = 1$ and $\text{Im}\left(\frac{z+1}{z+2}\right) = \frac{1}{5}$. Then the value of $|\text{Re}(\overline{z+2})|$ is

- (1) $\frac{2\sqrt{6}}{5}$ (2) $\frac{24}{5}$
(3) $\frac{1+\sqrt{6}}{5}$ (4) $\frac{\sqrt{6}}{5}$

Q63. If the set $R = \{(a, b) : a + 5b = 42, a, b \in \mathbb{N}\}$ has m elements and $\sum_{n=1}^m (1 - i^n) = x + iy$, where $i = \sqrt{-1}$, then the value of $m + x + y$ is

- (1) 12 (2) 4
(3) 8 (4) 5

Q64. If $\sin x = -\frac{3}{5}$, where $\pi < x < \frac{3\pi}{2}$, then $80(\tan^2 x - \cos x)$ is equal to

- (1) 108 (2) 109
(3) 18 (4) 19

Q65. The equations of two sides AB and AC of a triangle ABC are $4x + y = 14$ and $3x - 2y = 5$, respectively. The point $(2, -\frac{4}{3})$ divides the third side BC internally in the ratio 2 : 1. the equation of the side BC is

- (1) $x + 3y + 2 = 0$ (2) $x - 6y - 10 = 0$
 (3) $x - 3y - 6 = 0$ (4) $x + 6y + 6 = 0$

Q66. Let the circles $C_1 : (x - \alpha)^2 + (y - \beta)^2 = r_1^2$ and $C_2 : (x - 8)^2 + (y - \frac{15}{2})^2 = r_2^2$ touch each other externally at the point (6, 6). If the point (6, 6) divides the line segment joining the centres of the circles C_1 and C_2 internally in the ratio 2 : 1, then $(\alpha + \beta) + 4(r_1^2 + r_2^2)$ equals

- (1) 125 (2) 130
 (3) 110 (4) 145

Q67. Let $H : \frac{-x^2}{a^2} + \frac{y^2}{b^2} = 1$ be the hyperbola, whose eccentricity is $\sqrt{3}$ and the length of the latus rectum is $4\sqrt{3}$. Suppose the point $(\alpha, 6)$, $\alpha > 0$ lies on H . If β is the product of the focal distances of the point $(\alpha, 6)$, then $\alpha^2 + \beta$ is equal to

- (1) 172 (2) 171
 (3) 169 (4) 170

Q68. Let $A = \begin{bmatrix} 2 & a & 0 \\ 1 & 3 & 1 \\ 0 & 5 & b \end{bmatrix}$. If $A^3 = 4A^2 - A - 21I$, where I is the identity matrix of order 3×3 , then $2a + 3b$ is equal to

- (1) -9 (2) -13
 (3) -10 (4) -12

Q69. Let $[t]$ be the greatest integer less than or equal to t . Let A be the set of all prime factors of 2310 and $f : A \rightarrow \mathbb{Z}$ be the function $f(x) = \left\lceil \log_2 \left(x^2 + \left\lceil \frac{x^3}{5} \right\rceil \right) \right\rceil$. The number of one-to-one functions from A to the range of f is

- (1) 25 (2) 24
 (3) 20 (4) 120

Q70. For the function $f(x) = (\cos x) - x + 1$, $x \in \mathbb{R}$, between the following two statements (S1) $f(x) = 0$ for only one value of x in $[0, \pi]$. (S2) $f(x)$ is decreasing in $[0, \frac{\pi}{2}]$ and increasing in $[\frac{\pi}{2}, \pi]$.

- (1) Both (S1) and (S2) are correct. (2) Both (S1) and (S2) are incorrect.
 (3) Only (S2) is correct. (4) Only (S1) is correct.

Q71. Let $f(x) = 4 \cos^3 x + 3\sqrt{3} \cos^2 x - 10$. The number of points of local maxima of f in interval $(0, 2\pi)$ is

- (1) 3 (2) 4
 (3) 1 (4) 2

Q72. The number of critical points of the function $f(x) = (x - 2)^{2/3}(2x + 1)$ is

- (1) 1 (2) 2
 (3) 0 (4) 3

Q73. Let $I(x) = \int \frac{6}{\sin^2 x (1 - \cot x)^2} dx$. If $I(0) = 3$, then $I(\frac{\pi}{12})$ is equal to

(1) $2\sqrt{3}$

(2) $\sqrt{3}$

(3) $3\sqrt{3}$

(4) $6\sqrt{3}$

Q74. The value of $k \in \mathbb{N}$ for which the integral $I_n = \int_0^1 (1-x^k)^n dx, n \in \mathbb{N}$, satisfies $147I_{20} = 148I_{21}$ is

(1) 14

(2) 8

(3) 10

(4) 7

Q75. Let $f(x)$ be a positive function such that the area bounded by $y = f(x), y = 0$ from $x = 0$ to $x = a > 0$ is $e^{-a} + 4a^2 + a - 1$. Then the differential equation, whose general solution is $y = c_1 f(x) + c_2$, where c_1 and c_2 are arbitrary constants, is

(1) $(8e^x - 1) \frac{d^2y}{dx^2} + \frac{dy}{dx} = 0$

(2) $(8e^x - 1) \frac{d^2y}{dx^2} - \frac{dy}{dx} = 0$

(3) $(8e^x + 1) \frac{d^2y}{dx^2} - \frac{dy}{dx} = 0$

(4) $(8e^x + 1) \frac{d^2y}{dx^2} + \frac{dy}{dx} = 0$

Q76. Let $y = y(x)$ be the solution of the differential equation

$$(1 + y^2)e^{\tan x} dx + \cos^2 x (1 + e^{2 \tan x}) dy = 0, y(0) = 1. \text{ Then } y\left(\frac{\pi}{4}\right) \text{ is equal to}$$

(1) $\frac{2}{e}$

(2) $\frac{2}{e^2}$

(3) $\frac{1}{e}$

(4) $\frac{1}{e^2}$

Q77. The set of all α , for which the vectors $\vec{a} = \alpha \hat{i} + 6\hat{j} - 3\hat{k}$ and $\vec{b} = t\hat{i} - 2\hat{j} - 2\alpha t\hat{k}$ are inclined at an obtuse angle for all $t \in \mathbb{R}$, is

(1) $\left(-\frac{4}{3}, 1\right)$

(2) $[0, 1)$

(3) $\left(-\frac{4}{3}, 0\right]$

(4) $(-2, 0]$

Q78. If the shortest distance between the lines $L_1 : \vec{r} = (2 + \lambda)\hat{i} + (1 - 3\lambda)\hat{j} + (3 + 4\lambda)\hat{k}, \lambda \in \mathbb{R}$ is $\frac{m}{\sqrt{n}}$ and $L_2 : \vec{r} = 2(1 + \mu)\hat{i} + 3(1 + \mu)\hat{j} + (5 + \mu)\hat{k}, \mu \in \mathbb{R}$, where $\gcd(m, n) = 1$, then the value of $m + n$ equals

(1) 390

(2) 384

(3) 377

(4) 387

Q79. Let $P(x, y, z)$ be a point in the first octant, whose projection in the xy -plane is the point Q . Let $OP = \gamma$; the angle between OQ and the positive x -axis be θ ; and the angle between OP and the positive z -axis be ϕ , where O is the origin. Then the distance of P from the x -axis is

(1) $\gamma \sqrt{1 - \sin^2 \phi \cos^2 \theta}$

(2) $\gamma \sqrt{1 - \sin^2 \theta \cos^2 \phi}$

(3) $\gamma \sqrt{1 + \cos^2 \phi \sin^2 \theta}$

(4) $\gamma \sqrt{1 + \cos^2 \theta \sin^2 \phi}$

Q80. Let the sum of two positive integers be 24. If the probability, that their product is not less than $\frac{3}{4}$ times their greatest possible product, is $\frac{m}{n}$, where $\gcd(m, n) = 1$, then $n - m$ equals

(1) 10

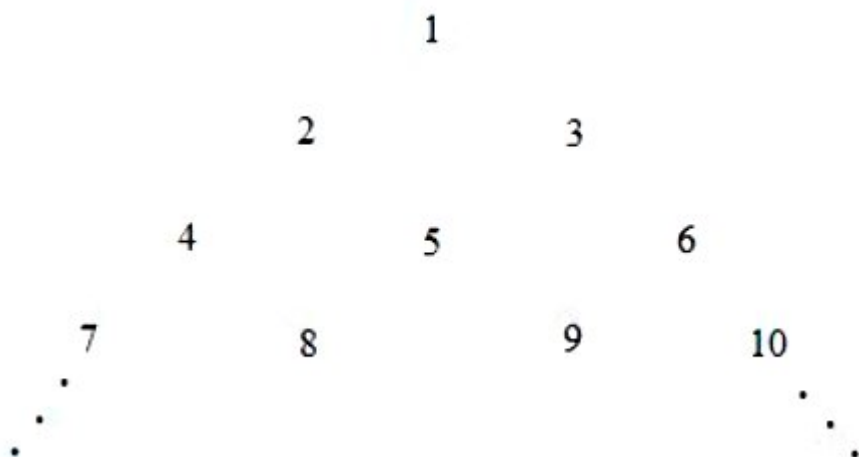
(2) 9

(3) 11

(4) 8

Q81. The number of 3-digit numbers, formed using the digits 2, 3, 4, 5 and 7, when the repetition of digits is not allowed, and which are not divisible by 3, is equal to _____

Q82. Let the positive integers be written in the form :



If the k^{th} row contains exactly k numbers for every natural number k , then the row in which the number 5310 will be, is _____

Q83. Let $\alpha = \sum_{r=0}^n (4r^2 + 2r + 1)^n C_r$ and $\beta = \left(\sum_{r=0}^n \frac{{}^nC_r}{r+1} \right) + \frac{1}{n+1}$. If $140 < \frac{2\alpha}{\beta} < 281$, then the value of n is _____

Q84. If the orthocentre of the triangle formed by the lines $2x + 3y - 1 = 0$, $x + 2y - 1 = 0$ and $ax + by - 1 = 0$, is the centroid of another triangle, whose circumcentre and orthocentre respectively are $(3, 4)$ and $(-6, -8)$, then the value of $|a - b|$ is _____

Q85. The value of $\lim_{x \rightarrow 0} 2 \left(\frac{1 - \cos x \sqrt{\cos 2x} \sqrt[3]{\cos 3x} \dots \sqrt[10]{\cos 10x}}{x^2} \right)$ is _____

Q86. Let $A = \begin{bmatrix} 2 & -1 \\ 1 & 1 \end{bmatrix}$. If the sum of the diagonal elements of A^{13} is 3^n , then n is equal to _____

Q87. If the range of $f(\theta) = \frac{\sin^4 \theta + 3 \cos^2 \theta}{\sin^4 \theta + \cos^2 \theta}$, $\theta \in \mathbb{R}$ is $[\alpha, \beta]$, then the sum of the infinite G.P., whose first term is 64 and the common ratio is $\frac{\alpha}{\beta}$, is equal to _____

Q88. Let the area of the region enclosed by the curve $y = \min\{\sin x, \cos x\}$ and the x axis between $x = -\pi$ to $x = \pi$ be A . Then A^2 is equal to _____

Q89. Let $\vec{a} = 9\hat{i} - 13\hat{j} + 25\hat{k}$, $\vec{b} = 3\hat{i} + 7\hat{j} - 13\hat{k}$ and $\vec{c} = 17\hat{i} - 2\hat{j} + \hat{k}$ be three given vectors. If $\vec{r}$ is a vector such that $\vec{r} \times \vec{a} = (\vec{b} + \vec{c}) \times \vec{a}$ and $\vec{r} \cdot (\vec{b} - \vec{c}) = 0$, then $\frac{|593\vec{r} + 67\vec{a}|^2}{(593)^2}$ is equal to _____

Q90. Three balls are drawn at random from a bag containing 5 blue and 4 yellow balls. Let the random variables X and Y respectively denote the number of blue and yellow balls. If $\bar{X}$ and $\bar{Y}$ are the means of X and Y respectively, then $7\bar{X} + 4\bar{Y}$ is equal to _____

ANSWER KEYS

1. (2)	2. (2)	3. (3)	4. (1)	5. (2)	6. (4)	7. (2)	8. (2)
9. (2)	10. (2)	11. (1)	12. (1)	13. (3)	14. (4)	15. (2)	16. (4)
17. (4)	18. (2)	19. (2)	20. (1)	21. (3)	22. (15)	23. (7)	24. (16)
25. (12)	26. (748)	27. (4)	28. (3)	29. (6)	30. (156)	31. (3)	32. (3)
33. (2)	34. (1)	35. (1)	36. (4)	37. (4)	38. (1)	39. (3)	40. (2)
41. (4)	42. (3)	43. (4)	44. (2)	45. (2)	46. (1)	47. (4)	48. (1)
49. (2)	50. (1)	51. (5)	52. (6)	53. (55)	54. (32)	55. (5)	56. (1)
57. (0)	58. (5)	59. (591)	60. (5)	61. (1)	62. (1)	63. (1)	64. (2)
65. (1)	66. (2)	67. (2)	68. (2)	69. (4)	70. (4)	71. (4)	72. (2)
73. (3)	74. (4)	75. (4)	76. (3)	77. (3)	78. (4)	79. (1)	80. (1)
81. (36)	82. (103)	83. (5)	84. (16)	85. (55)	86. (7)	87. (96)	88. (16)
89. (569)	90. (17)						